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REGULAR PAPER

A survey on Urban spatial computing system

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Abstract
As a type of information infrastructure, the spatial information system (SIS) can be used to store, process, and analyze geospatial data. The SIS-based urban spatial computing system (USCS) for multi-field urban applications is a critical platform for collecting and analyzing various heterogeneous big data in the city. However, the sources of city-related data are diverse and complex, making it hard to achieve comprehensive data integration and management in USCS. Simultaneously, the technological architecture of USCS is also evolving, requiring a holistic review of USCS-related theory and technology. This paper reviewed multiple aspects related to USCS based on existing findings, and the main points observed are as follows: (a) various data types along with their sources and acquisition methods; (b) a comparative analysis of multiple data models and database systems; (c) an elaboration of USCS application services and frontier technologies driving USCS evolution; d) discussion of typical USCS, accompanied by the presentation of a USCS architectural design proposal for reference. The review provides relevant researchers with a quick and comprehensive perspective on USCS and promotes their research in some frontier directions.

Keywords Spatial information · Data management · Computing system · System architecture design

Mathematics Subject Classification 68 Computer Science · 38U35 Computing methodologies for information systems (hypertext navigation, interfaces, decision support, etc.)

Extended author information available on the last page of the article

1 Introduction

Nowadays, the promotion and applications of spatial information system (SIS) have gradually become widespread and have played a significant role in multiple fields [1]. Spatial data reflects the spatial distribution characteristics of geographic entities, including their position and shape, as well as the spatial relationships between them [2]. Correspondingly, an SIS revolving around multi-type spatial data, covering the entire workflow from data acquisition and processing to analysis. Driven by urban informatics and extensive practical needs, urban spatial computing systems (USCS) are gradually playing an important role. As a type of SIS, USCS has a broad range of applications in urban areas, spanning from urban construction to social services. Figure 1 illustrates a concise overview of the application case of USCS, highlighting three primary layers: data aggregation of urban spatial data, system design and development, and the provision of services to various industries or fields.

As the principal domains of human habitation and livelihood, urban spaces usually have complex scenarios and involve diverse data types and formats, resulting in the development of USCS encountering considerable challenges that set them apart from other SIS [3]. Furthermore, the large population residing in cities, comprising more than half of the total population, along with their significant political, economic, and cultural status, all contribute to the significance and indispensability of USCS [4]. The wide-ranging and diverse data sources in urban space present a substantial challenge in data acquisition, organization, management, and analysis [5, 6]. Besides, the construction of smart cities, advancements in various technical fields, and the need for intricate applications all contribute to the growing complexity and difficulty in the research and development of USCS [7].

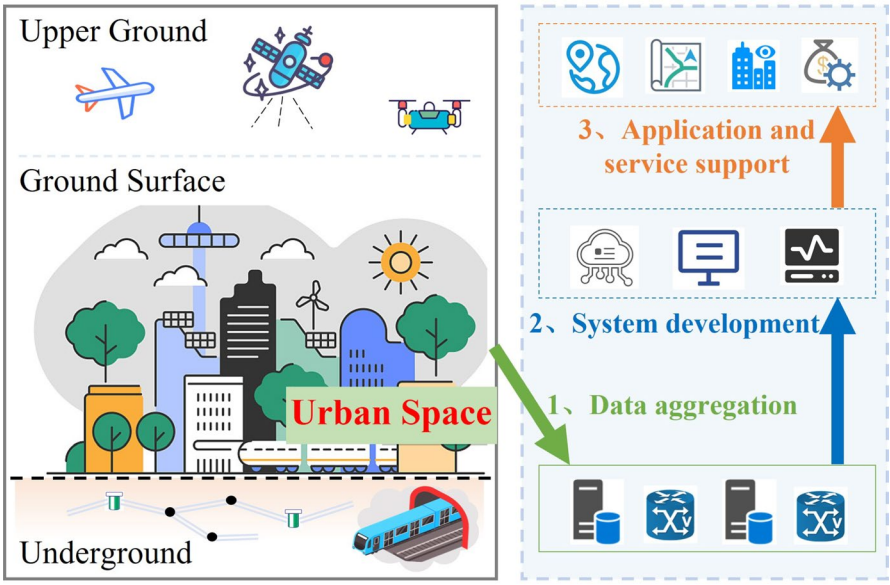


Fig. 1 Overview of the application of urban spatial computing systems

The cutting-edge theory and technology framework of USCS has been continually evolving alongside the advancements in technologies such as artificial intelligence (AI) technology and the Internet of Things (IoT) [8]. At a macro level, the evolution of USCS is evident in comprehensive transformations spanning data acquisition methods, storage technologies, processing techniques, and service modalities [9, 10]. The diversification of data acquisition, propelled by IoT advancements, significantly amplifies data richness in urban environments [11]. Concurrently, big data technologies advance the efficiency of managing massive datasets. The synergy of AI technologies, such as deep learning and cloud computing, notably elevates data processing capabilities [12]. The pervasive application across domains, coupled with escalating demands for human–machine interaction and high availability, perpetuates the continual evolution of USCS.

Although there are some reviews of urban information systems, they are often oriented to a single industry or application field, such as the research in the transportation or energy field by Qureshi et al [13, 14]. Besides, these reviews have been relatively narrow in scope and fail to encompass the full range of data types and related theories, as they are tailored to specific industry applications [15–17]. Overall, the current review research on USCS is lacking in a holistic summary of the relevant theories and technical frameworks. In particular, there is little discussion of the data sources and associated management methods. This paper focuses predominantly on the theme of USCS, thoroughly discussing the acquisition, organization, management, and application of urban spatial data. In addition, several frontier technologies related to urban spatial information processing and applications were discussed. Meanwhile, we analyzed some USCS systems, proposed a system architecture design, and gave a system demonstration. By deeply discussing the current developmental status of technologies and applications related to USCS, the paper serves as a vital reference and practical guide for research on USCS, geographic information system (GIS), and other professionals engaged in urban spatial data research. The main contributions of this paper are as follows:

1. Drawing upon the relevant literature, we provide detailed insights into the reams of data sources and data management constituting the foundational components of USCS. This paper delves into various aspects including data category, data acquisition, data model, and database system related to urban space, effectively assisting researchers in understanding the theory related to urban spatial data and data management.
2. We elaborate on two angles of frontier technology and application requirements to reveal the future development direction of USCS. It discusses the impact of several frontier technology fields on the development and technological evolution of USCS, along with categorizing a wide range of services and multi-field applications in urban settings with the support of USCS.
3. We conduct a comparative analysis of existing USCS to unveil the current development and application status. Meanwhile, we propose a novel USCS architecture design and showcase a system implementation case, inspiring researchers in terms of the design and development of USCS.

This paper explores the theories, mainstream technologies and literature related to USCS, and is committed to making an in-depth and comprehensive analysis of its evolution and current status to guide researchers in their research directions. In terms of literature, we cite references that are representative and most credible in demonstrating the latest theories and mainstream technologies. To achieve this goal, we have rationally arranged the structure of the paper, and the following sections are organized as shown in Fig. 2. Section 2 presents the data sources in urban spatial scenes, including data category and data acquisition. Section 3 gives the technical state of data management, including the data model and database system. Section 4 discussed several main frontier technologies that have the greatest impact on USCS. Section 5 discusses the various applications and services that USCS can provide. Section 6 illustrates several existing USCS platforms. We propose an architecture design scheme for USCS in Sect. 7. Future directions are given in Sect. 8 and conclusions are drawn in Sect. 9.

2 Data sources

As the input and processing object of USCS, the data source affects the system functions and service capabilities. Therefore, we first discuss the data sources that are closely related to USCS. This paper delved into two dimensions of data in urban spaces: data categories and data acquisition. The data category section sorted out multiple types of data widely existing in urban spaces, while the data acquisition section listed various data acquisition methods and data characteristics in different urban scenarios.

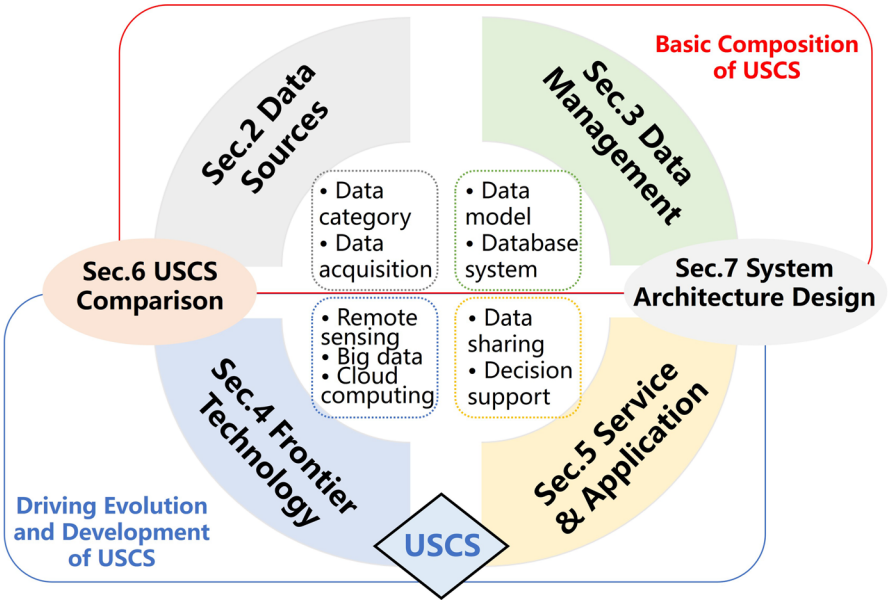


Fig. 2 The overall architecture and research contents of this article

2.1 Data category

A city is a spatial and territorial system centered around its inhabitants, characterized by efficient space utilization, and aimed at achieving agglomeration effects [18]. Various industries and application domains define urban information based on its categories, distribution, patterns, and other characteristics, leading to different classification criteria. Broadly, urban data with various types and sources can generally be divided into three categories: geographic information, economic entity, and social event [19]. Among them, geographical information is an important part of urban information resources, with high versatility and significant sharing demands. Geographic information mainly includes natural environment data consisting of elements such as water, landforms, vegetation, atmosphere, rocks, and minerals, as well as artificial environment data covering elements such as buildings, public infrastructure, transportation, and functional areas [20]. Besides, economic entity data can be divided into government-related data, enterprise-related data, and personal-related data at a macro level, while social event data includes various types of data related to urban disasters, environmental governance, emergency rescue and other events [21].

Urban data possesses characteristics of multiple spatial scales, temporal scales, user objects, and thematic types [22]. Figure 3 illustrates the four-dimensional attributes of urban data and provides specific examples for each attribute. Addressing the challenges of urban data diversification and coordination and establishing an application-driven urban data conceptual model from these four dimensions are fundamental requirements to construct a highly efficient USCS. During the construction of the urban spatial information database, these four dimensions can be treated as separate dimension tables, establishing logical associations and integrated storage for each data, thereby facilitating the cooperative analysis of urban data from various dimensions [23]. Firstly, spatial and temporal attributes are the essential elements of urban data, and these data have apparent spatiotemporal integration characteristics. The overall management of spatiotemporal attributes can serve as a foundation for the construction and effective operation of USCS. Furthermore, urban data exhibit characteristics of multiple thematic types, which may reflect different connotations and characteristics in different applications. Due to the characteristics of multiple thematic types, urban data also exhibit characteristics of multi-user objects.

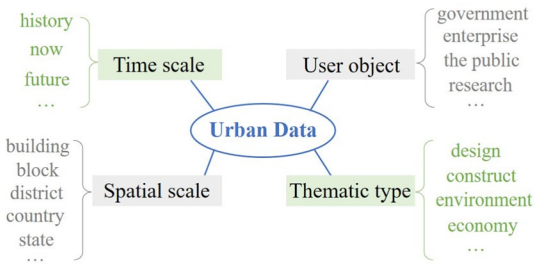


Fig. 3 Four-dimensional characteristics of urban data

2.2 Data acquisition

With the coordinated development of various technologies, the methods of obtaining data in urban space have become diverse, and the quantity of city-related data is greater, and the categories are more extensive. Currently, prevalent data acquisition methods around cities include remote sensing (RS), IoT, video surveillance, navigation and positioning, etc [24]. Each method has different observation objects, and the data collected varies in volume and type [25]. We sorted out the acquisition or observation objects of each method and summarized the characteristics of the corresponding data. The detailed description and comparison are shown in Table 1, which specifies the type and characteristics of the data obtained by each method.

Table 1 Comparison of urban data acquisition method

Acquisition method	Collection object	Data characteristics
Remote Sensing	The Earth's surface, including the mainland and oceans, as well as specific targets on land identified through aerial remote sensing	The volume of data is large, the types are diverse, and the formats are varied. The cost of data transmission and processing is relatively high, but valuable data information can be extracted for multiple fields [26]. High spatial resolution is accompanied by low temporal resolution; real-time processing is insufficient [27]. Common data types: raster, video, point cloud, etc
Internet of Things	According to the different sensor devices, information such as sound, light, electricity, heat, pressure, biology, chemistry, and location can be collected	The information content and format captured by different sensors are different, and the data collected generally have strong real-time characteristics [24]. High-frequency continuous acquisition and wide spatial coverage; edge computing can improve real-time processing efficiency [28]. Common data types: JSON, XML, CSV, JPG, PNG, MP4, HDF5, etc
Video Surveillance	It can achieve detection and monitoring region of interest and is widely used in various urban scenarios, such as road intersections, shopping malls, etc	It is able to display and record on-site information in real-time, and the data features are intuitive, accurate, timely, and rich in content. High temporal resolution and spatial resolution determine the picture clarity; suitable for real-time scenes of dynamic monitoring [25]. Common data types: MP4, MKV, etc
Navigation Positioning	The various moving objects in the city, such as people and vehicles, reflect the urban changes and are a direct embodiment of the city's dynamism	The position information obtained from terminals such as Beidou and GPS can be used for high-precision continuous indoor and outdoor positioning and real-time navigation. High temporal and spatial resolution requires fast processing and immediate feedback [29]. Common data types: GPX, KML, JSON, CSV, etc
Site Monitoring	Conduct targeted, timed, quantitative, and continuous observations around meteorology, water quality, air pollution, and more	The continuous attribute values of specific objects are generally data in key-value format, and the structure is relatively simple. High time resolution, real-time or near real-time continuous data acquisition [30]. Common data types: NetCDF, CSV, JSON, XML, etc
Field Measurement	Regarding various types of objects in the field, it is possible to obtain information on the location, shape, and attributes of natural and social elements	This type of data is usually text or simple relational statistical data, with multiple categories and varying data volumes. Low temporal resolution, spatial resolution is affected by terrain, and weak real-time processing capability [28]. Common data types: GeoJSON, SHP, KML, GML, Excel, CSV, etc

Among the various methods for acquiring urban data, RS observation is a significant data source [26]. There are currently thousands of satellites in orbit around the Earth, as well as various irregular aviation, unmanned aerial vehicles (UAV), and ground RS operations, all of which are conducting continuous observations and data acquisition of the Earth's surface, including urban spaces [27]. Furthermore, with the development of sensing devices such as visible light photography, color infrared photography, thermal infrared scanning, and multispectral scanning, people can access urban remote sensing data with multiple resolutions, spectral bands, and quality levels [30]. The characteristics of remote sensing data, namely a high data volume and complex processing, are further compounded by the current trend of collaborative data processing, which presents a significant challenge in the context of remote sensing data analysis. Researchers can obtain valuable information from massive remote sensing data and apply it to many fields, such as urban land use, urban expansion, environmental protection, urban governance, etc. In the future, remote sensing observations will play a significant role and more effectively assist urban planning, smart city construction, disaster relief and reduction, environmental governance and sustainable urban development [31].

In addition to remote sensing observation methods, some other data acquisition methods can also obtain large-volume and diverse urban data. As for data acquisition methods such as IoT, navigation and positioning, and video surveillance, real-time and abundant data on specific targets in the city can be obtained [28]. For example, real-time dynamic data of terminals, vehicle trajectory data, road monitoring data, etc. Meanwhile, the data acquisition methods of field measurement and site observation can obtain extensive information on natural and social elements, effectively supporting in-depth development and application in various fields [29]. These data acquisition methods mainly collect associated data of point, line, and surface type target objects in the city, which are generally vector data or statistical data surrounding specific attributes. In short, the current diverse data acquisition methods have caused the rapid growth of urban data volume, making it hard to develop an efficient USCS to manage, process, and utilize these data.

3 Data management

Data management encompasses the systematic collection, storage, processing, and utilization of data to extract its full potential. Concurrently, researches and applications within the field of USCS seek to leverage data management techniques to cater to a diverse array of urban-related endeavors. To fulfill this objective, it becomes imperative to initially structure and integrate heterogeneous urban data from multiple sources, employing suitable data models. This section undertakes an in-depth examination of various prevalent data models and database systems, providing a nuanced analysis of their individual attributes. By doing so, it endeavors to furnish USCS researchers with both theoretical insights and technical guidance.

3.1 Data model

The data model serves as an abstract representation of real-world data attributes, furnishing a framework for database system information representation and operations. It stands as the fundamental core of the database system. Spatial data, among diverse urban data types, poses distinct challenges in digital representation compared to relational statistics and text data [32].

In the domain of spatial data, vector and raster data types are the predominant forms of representation and transmission, exerting a considerable influence within this field. Furthermore, a multitude of data models are in existence, each displaying distinctive characteristics and suitability for specific contexts and scenarios [33]. In summary, these multiple data models, designed for the representation of spatial data, underpin the essential foundation for the exchange and utilization of spatial data. Subsequently, we will explore prevalent data models to help researchers use appropriate data models to represent spatial data.

(1) Raster Data Model

The raster data model comprises regular square or rectangular grids, wherein each grid or pixel is uniquely identified by its corresponding row and column coordinates, and the associated value signifies the attribute associated with the depicted content [34]. The interpretation of these grid points is user-customized, affording considerable versatility, often leveraging empirically measured values or numerical codes denoting specific categories. Figure 4 (a) provides an elucidative instance of raster data, wherein each grid point corresponds to a tangible geographical area on the Earth's surface. Decreasing the size of grid cells enhances data precision but simultaneously engenders larger data volumes [35]. With heightened resolution, the storage demands for raster data exhibit an exponential increase.

The utilization of raster data as a means of recording and storing continuously evolving geographic spatial information has proven to be particularly well-suited, especially in regions characterized by substantial variations in attributes across space. Diverse data formats, encompassing satellite imagery, an extensive array of

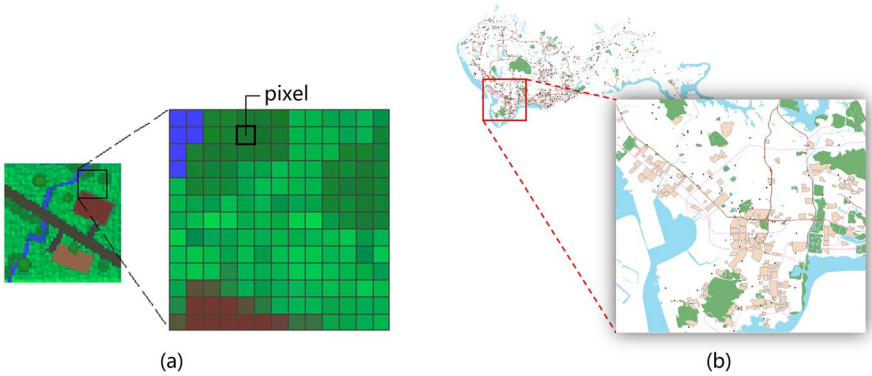


Fig. 4 Examples of raster data and vector data

remote sensing data products, and maps illustrating specific urban characteristics, routinely adopt raster data model. This approach not only fosters efficient computational processing but also bolsters the seamless exchange and dissemination of data [36]. Nevertheless, the universal applicability of the raster data model is not absolute when considering all types of spatial data. It is constrained by limitations, particularly in the representation of point and linear entities, as well as in the articulation and analysis of spatial relationships among geographic entities.

(2) Vector Data Model

The vector data model affords an efficient means of representing spatial entities and their intricate interconnections, all while conserving storage space. This data model excels in the portrayal of cartographic elements, ensuring a high degree of precision in delineating geographical spatial objects [37]. Typically, each geographical entity encompasses a multitude of attributes, including but not limited to name, location, population, and economic output. The vector data model exhibits a remarkable proficiency in capturing both the positional coordinates and a diverse array of associated attributes for these entities [38]. Within a two-dimensional spatial context, urban geographical entities can be classified into three distinct classes: point entities, line entities, and polygon entities [39]. Similarly, vector data can be categorized into three fundamental types, namely points, lines, and polygons. Specifically, vector points such as hospitals and schools, vector lines such as rivers and roads, and vector polygons of closed areas such as buildings and lakes.

Utilizing vector data models for the representation of geographic data offers enhanced precision while concurrently reducing data volume. This, in turn, enables a comprehensive depiction of urban spatial relationships and the efficient recording of object attributes [40]. The data model demonstrates exceptional proficiency in the realms of network transmission, retrieval, and analytical operations [41]. Nevertheless, its complexity increases significantly when used in overlay analyses. As depicted in Fig. 4 (b), we provide illustrative instances of various vector data types, highlighting specific examples such as green spaces, metro routes, and dining establishments within Shenzhen city.

(3) Vector-raster integration model

The vector and raster data models each possess distinctive merits and limitations. However, when harmoniously amalgamated, they offer a more holistic means of recording and representing urban spatial features. The integration of vector and raster paradigms results in a cohesive data structure that effectively encapsulates the intrinsic attributes of both data models [42]. On one hand, it retains all the essential characteristics of vector data, including well-defined positional information and the ability to establish topological relationships [43]. On the other hand, each raster cell along the path is directly associated with the object. In essence, the vector-raster integrated data model predominantly operates within a raster framework, employing the technique of filling linear object paths and spatial object areas as the foundational approach for this integrated data structure [44].

(4) Network data model

The network data model functions as an abstract representation of tangible network systems found in domains such as transportation, communication, and pipeline networks [45]. It consists of two essential components: edges, which symbolize linear entities and operate as conduits for communication or resource conveyance, and nodes, which function as termination points for edges or points of intersection between them. Nodes effectively represent transfer hubs, intersections, and analogous features. While the network data model adeptly encapsulates the topological interconnections between diverse target nodes and pathways, it does not incorporate the exact geographical coordinates of physical objects [46]. The network data model can be employed to represent a multitude of data sets exhibiting intricate relationships and interconnections within urban environments.

(5) Spatiotemporal data model

Within spatiotemporal data models, a temporal dimension is integrated alongside the spatial and attribute dimensions. This comprehensive data model adeptly captures and manages dynamic alterations within spatial entities and their associated attribute relationships across time [47, 48]. The semantic richness inherent in the spatiotemporal data model empowers a more precise emulation of the intricacies of the real world [49]. In light of the exigent need for processing spatiotemporal data and extracting valuable insights from it, this data organization paradigm holds immense promise and potential for diverse applications [50].

One prominent instantiation of such a data model, extensively embraced in both theoretical frameworks and practical applications, is the Data Cube [51]. In the realm of remote sensing image data organization, the Data Cube Organization paradigm is prominently recognized for its characteristic approach in structuring remote sensing Analysis-Ready Data (ARD) into extensive gridded multidimensional datasets, complemented by systematic indexing across various dimensions or facets. This organizational framework adeptly employs spatial partitioning techniques to segment analysis-ready datasets into well-defined, non-overlapping tiles composed of gridded data during the initial data ingestion phase. Subsequently, based on the timestamp of data capture, these gridded data tiles are amalgamated and presented in a coherent cubic formation, subsequently encapsulated into multidimensional data files or accommodated within array databases for storage and retrieval purposes.

(6) Object-oriented data model

The data model interprets spatial entities as assemblies of objects, enabling the seamless amalgamation and manipulation of spatial geometric data and attribute data within a unified entity. This conceptual framework establishes a congruence between the two aspects of data [52]. Consequently, this data model endows a multitude of expressive capabilities, affording a more intuitive representation of the physical world [53]. Notably, it excels in the representation of spatial entities that encompass multimedia data, as it encapsulates spatial entities alongside their associated images,

videos, audio recordings, and supplementary information, thereby affording a comprehensive portrayal of spatial phenomena [54].

(7) Three-dimensional data model

Urban environments contain a wide array of data types that extend beyond two-dimensional representations, encompassing information about buildings, pipeline models, and other three-dimensional (3D) spatial entities [55]. To adequately capture and express these complex structures, the deployment of corresponding 3D data models becomes imperative. Broadly, these data models can be categorized into two primary classifications. The first classification utilizes surface-based representations, such as grid structures and triangulated irregular networks (TINs), with a primary emphasis on the portrayal of 3D surfaces, encompassing terrains, geological features, and various other spatial surfaces. The second classification embraces volume-based representations, typified by octrees, employing volumetric topology as their fundamental representation methodology.

Contemporary urban environments present an intricate tapestry of spatial scales within 3D models, encompassing expansive geographic representations to minutely detailed indoor renderings, thereby posing substantial challenges in ensuring uniformity across multi-scale 3D models [56, 57]. Conventional 3D data models have predominantly been tailored to cater to specific professional domains and applications [58]. Effectively representing 3D models across diverse urban contexts necessitates targeted research aimed at achieving cohesive articulation and representation of entities spanning various scales, spanning both above-ground and subterranean realms, in addition to indoor and outdoor environments.

In the multi-source heterogeneous data modeling of urban space, different data models can adapt to diverse data types and application scenarios because of their characteristics. For example, the raster data model is suitable for the digital representation of continuously distributed geographic phenomena, while the vector data model is good at processing precise geometric objects, including roads, rivers, points of interest, etc. Besides modeling buildings and terrain, three-dimensional data models are also particularly important in the management of urban underground pipeline networks. The spatiotemporal data model can integrate the time and space dimensions. Currently, there are some spatiotemporal big data platforms that combine real-life three-dimensional with natural resource services to support major project site selection and disaster warning, and improve urban governance efficiency. Network data models are often used to optimize traffic flow and realize dynamic scheduling in combination with IoT data. Object-oriented data models are suitable for complex entity modeling, such as associating building models with GIS data and supporting three-dimensional spatial analysis. During the implementation process, it is necessary to research the characteristics of the data and the application scenarios.

3.2 Database system

In the context of various urban spatial data, the imperative of implementing a robust database system becomes evident, as it plays an integral role in facilitating com-

prehensive data storage management. This, in turn, serves as the cornerstone for delivering multiple data service support to applications situated within urban-related domains [59]. Database systems are explicitly engineered to deal with the massive data and are specifically tailored to offer an effective framework for both the preservation and retrieval of information within the database [60].

Nevertheless, the landscape of urban-related data is characterized by its inherent diversity, wherein each unique data type exhibits distinct attributes necessitating the deployment of an associated database system to fulfill its storage requirements. We conducted a comprehensive comparative analysis of diverse data management solutions and provided an in-depth exposition of the distinctive attributes inherent to each database system. A detailed presentation of the specific findings is presented in Table 2. In the subsequent section, we will expounding the categories of databases, accompanied by an elucidation of select representative and well-established database systems.

(1) Categories of Database System

Table 2 Data organization-storage solution overview

Category	Name		Characteristics Description of Database Systems
Relational database	MySQL		Small size, fast speed, open source, support multiple operating systems; high read-write performance and horizontal scalability; does not support custom data types and hot backup
	SQL Server		Easy-to-use, more flexible data management and analysis; only supports Windows system, no security certificate; support XML, spatial data types
	PostgreSQL		Open source, free, mature community, powerful plug-in tools, support for nesting; flexible vertical expansion and support for custom types
	Oracle		Support all mainstream platforms, good performance, relatively complex operation, expensive, high maintenance costs; compatible with multiple platforms
NoSQL	MongoDB		Document-type database, written in C++ language, has high efficiency in processing web page data; high write throughput and support for sharded cluster expansion
	Cassandra		High scalability, no single point of failure, flexible data storage; high write throughput, supports limited data types
	RabbitMQ		Provide unified messaging service and message-oriented middleware design; support mirror queue expansion
	Distributed database	Redis	Memory-based key-value NoSQL database, fast; provides rich data structures and supports data persistence; excellent read and write performance; supports multiple data types
		HBase	Column-oriented, highly reliable, and scalable, which can solve the sparsity problem and has strong scalability; suitable for PB-level sparse data
		Ceph	Support file storage, object storage, block storage, strong consistency, and decentralization; support EB-level expansion
		Neo4J	A graph database with high-performance capabilities, storing relational data of knowledge graph; open source, implemented based on Java, and supports transactions
		ES	Support distributed search engine based on Lucene inverted index, with high stability; good performance, easy to use; near real-time search

In light of data storage attributes, databases can be classified into two classes: relational and non-relational databases. Moreover, there exist file systems that facilitate file-centric archival and storage mechanisms.

1) Relational Database

The relational database is founded upon the principles of the relational data model, which employs a tabular structure comprising rows and columns to organize data [61]. Within the framework of a relational database, entities and their interconnections are expressed through a unified structure consisting of multiple two-dimensional tables. Notable examples of contemporary relational databases encompass Oracle, SQL Server, PostgreSQL, MySQL, and others, all of which effectively handle statistical data associated with municipalities and aggregated data derived from diverse departments. Relational databases demonstrate particular prowess in preserving data consistency, integrity, and security [62]. However, they are ill-suited for the organization and management of unstructured data, especially when confronted with searching and establishing connections within extensive datasets [63]. Some relational databases offer extensibility through plugins designed to augment their storage capabilities, exemplified by the PostGIS component provided by PostgreSQL [64].

2) Non-relational Database

The rapid growth of unstructured and semi-structured data has led to a significant increase in their prevalence within datasets [65]. Conventional relational databases are inadequate for the efficient organization and management of this multifarious data [66]. Significantly, diverse data formats such as vector features, raster data, geographical terrain data, as well as multimedia content, all demand the adoption of non-relational databases as the optimal solution for the purpose of data organization and storage.

Non-relational databases, often denoted as NoSQL, encompass diverse database management systems that diverge from the conventional relational database paradigm [67]. They exhibit compatibility with database systems suitable for structured, unstructured, and other polymorphic data. Mainstream NoSQL databases include key-value stores, column stores, object stores, graph databases, document stores, and full-text search engines [68]. These offerings distinguish themselves from their relational counterparts by affording versatile storage capabilities [69]. NoSQL databases find widespread utility in the realm of big data processing, particularly when confronted with the challenge of managing extensive volumes of data characterized by a diversity of types and complex formats [70]. NoSQL databases have certain constraints, encompassing the absence of standardized query languages, reduced efficiency in complex queries, inconsistent data retrieval, and a comparably less mature user community when contrasted with the SQL database community [71].

3) File system

In the realm of urban multi-source data storage, file systems also offer solution for integrating data information. Although both file systems and database systems are enlisted for the purpose of data administration, they are driven by distinct design objectives and employ disparate implementation methodologies [72]. File systems are primarily engineered to structure and retain files on storage media, furnishing essential operations such as file creation, deletion, copying, and relocation. In contrast to database systems, which prioritize the management of data in a more holistic

sense, file systems adopt a file-centric approach with a primary focus on facilitating expeditious access to individual files. In the contemporary landscape, file systems find extensive utility in supporting data storage and management within distributed environments, with notable solutions including Gluster File System, Hadoop Distributed File System (HDFS), and Lustre.

(2) Typical Cases of Database System

With the proliferation of diverse data acquisition methods, the demand for massive data storage and processing has accelerated the development and application of distributed database systems [73]. We will summarize several leading distributed data storage systems and analyze their distinct characteristics, along with the data types they accommodate.

1) Redis

Redis, a memory-based NoSQL database, uses the key-value data model [74]. Within this framework, Redis efficiently structures data into key-value pairs, where each key serves as the unique symbol for retrieval, while the corresponding value signifies the authoritative data set [75]. Redis exhibits exceptional flexibility, serving as a versatile data structure server capable of accommodating an array of data types, including but not limited to strings, hashes, lists, sets, and sorted sets.

2) HBase

HBase makes up a distributed columnar database, underpinned by the HDFS. It is classified within the realm of NoSQL databases, characterized by its provision of primary attributes, including column-oriented storage, exceptional performance, scalability, and a heightened level of reliability [76]. Notably, HBase remains open-source and continues to serve external users effectively. The efficacy of HBase in accommodating loosely structured data engenders a remarkable versatility in columnar storage, liberating users from the obligation of enforcing uniformity across all rows during the process of adding new data [77].

3) Ceph

Ceph, a highly scalable, fault-tolerant, and high-performance distributed object storage system, offers remarkable versatility for accommodating extensive infrastructure needs [78]. The decentralized architectural approach inherent to Ceph systematically mitigates the vulnerabilities associated with single points of failure. As a result, Ceph excels in managing substantial datasets and effectively handling copious fragmented small files [79, 80]. Ceph offers diverse client interfaces, making it a common choice for serving as the foundational storage layer in cloud and container platforms.

4) Elasticsearch

Elasticsearch (ES) is a Java-based search server that employs RESTful interfaces for communication [81]. It is grounded in the inverted index method of Lucene and boasts capabilities for distributed deployment, rendering it a highly efficient and scalable search engine platform [82]. The full-text search engine furnished by ES is extensively applied in diverse domains. Prominent attributes of ES encompass its exceptional stability, feasibility, and performance, which collectively deliver a seamless and expeditious retrieval experience [83]. ES assumes a pivotal role as an indispensable search engine service for Internet cloud platforms.

5) Neo4J

Neo4j is distinguished by its adept utilization of graph modeling techniques for the organization and storage of structured data. The adoption of a graph-centric data model in Neo4j not only affords the capacity to store attributes associated with data entities but also lends itself to the proficient management and preservation of inter-entity relationships [84]. Neo4j enables the realization of remarkably efficient graph search and traversal methodologies, thus facilitating the effortless execution of data association queries and insertion operations [85].

6) Hadoop Distributed File System

HDFS is designed for the management of large-scale datasets. It distinguishes itself with robust fault tolerance mechanisms and the ability to enable efficient high-throughput data retrieval [86]. The architectural underpinnings of HDFS prioritize the efficient processing of large data batches. In its capacity as a foundational element within the open-source Hadoop distributed computing framework, HDFS assumes a pivotal role in the establishment of HBase databases and provides critical support for the execution of MapReduce distributed computing tasks [87].

In accordance with the specific attributes of data and the access patterns, the selection of an appropriate data management solution becomes imperative. Notably, when confronted with the challenge of handling extensive volumes of data, traditional database systems often prove inadequate in achieving efficient management. Therefore, within a distributed architecture framework, alternatives such as distributed file systems and distributed database systems emerge as viable options. For instance, HDFS proves invaluable for the storage and administration of extensive large files, graph databases excel in structuring data with intricate entity-to-entity relationships, and object storage systems offer an adept means of archiving substantial datasets characterized by numerous small files and varying data volumes. In addition, different database systems can meet diverse needs by extending their functions, especially in real-time data management. For example, PostgreSQL extends its geographic spatial functionality through the PostGIS plug-in, and in its ecosystem, it implements support for traditional GIS applications to support diversified processing of spatiotemporal data based on a wealth of extensions. In summary, for multi-source and multi-type data in urban spaces, appropriate data management solutions can be selected based on specific usage scenarios and data characteristics.

4 Frontier technologies driving USCS

The evolution of USCS has been significantly influenced by the advent of frontier technologies, including big data, cloud computing, and artificial intelligence. Furthermore, the urban observations approach, which is based on RS theory and technology, has yielded a wealth of invaluable information conducive to urban development and construction. This paper primarily discusses the impact of four frontier technologies, including remote sensing, on the evolution of USCS.

Within the realm of technology, distinct domains play unique and pivotal roles in shaping the evolution of USCS. As depicted in Fig. 5, the big data technology domain significantly bolsters the USCS by enhancing its capabilities for vast data storage and processing. Conversely, cloud computing domain elevates the USCS by

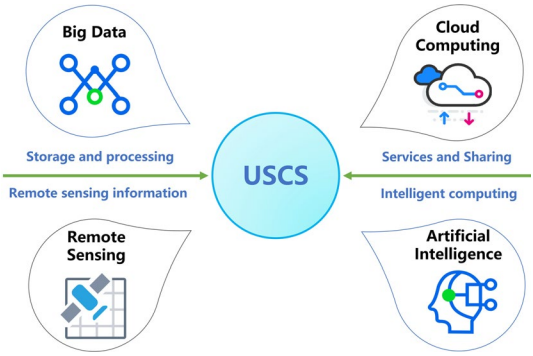


Fig. 5 Frontier technologies driving the development of USCS

facilitating robust system services and resource sharing. Furthermore, the domain of RS contributes to the enrichment of the USCS through the infusion of an abundance of knowledge-laden data. Simultaneously, AI domain empowers the USCS with sophisticated processing capabilities, imparting it with the potential for intelligent decision-making. Subsequently, a comprehensive examination of these frontier technologies' development and the transformative impacts they have engendered will be undertaken.

4.1 Impact of remote sensing on USCS

RS stands as the cornerstone of spatial exploration, enabling the harnessing of sensor technology to capture electromagnetic wave data emanating from distant terrestrial targets through radiation or reflection [88]. Over the past six decades, RS satellites have proliferated on a global scale, serving military, civilian, and commercial purposes, continually bolstering their detection capabilities and broadening the scope of applications. Presently, RS has found extensive applications across diverse domains, encompassing geological disaster forecasting, ecological environment assessment and conservation, land and mineral resource monitoring and administration, as well as the construction of national defense infrastructure, among others [89]. Of particular significance in urban contexts, RS plays an irreplaceable role, effectively supporting numerous activities related to urban governance, planning, and evaluation.

Nowadays, scholars across multiple disciplines and departments are dedicated to harnessing valuable insights from RS data to bolster urban development and construction [90]. On one hand, RS data enriches USCS with information and knowledge of limitless practical utility. However, this enrichment necessitates the integration of advanced RS data processing and analysis algorithms to derive copious valuable information from data and models, thereby facilitating applications across diverse domains. On the other hand, confronted with the vast volume of Earth observation data, we must undertake further research in the areas of storage management and efficient data processing to fully exploit the potential of RS data [91]. Looking ahead, the application of RS observation technology in urban contexts will continue to deepen, with its reach expanding even further.

4.2 Impact of big data on USCS

Big data refers to large and complex datasets that are challenging to process with traditional methods. It is characterized by high variety, increasing volume, and rapid velocity [92]. Big data technology represents a novel generation of technology and architecture engineered to derive value from vast and diverse data repositories through rapid data collection, discovery, and cost-effective analysis. In the contemporary landscape, big data plays an critical role in urban management, particularly within domains such as intelligent transportation, environmental monitoring, urban planning, and security.

Big data technology has exerted a profound and extensive influence on USCS, with notable implications in the realms of data storage, high-performance computation, and data processing and analysis. Firstly, it empowers the efficient administration of vast and diverse urban multi-source data, encompassing structured, semi-structured, and unstructured data formats [93, 94]. Secondly, it facilitates a spectrum of computational approaches, including batch processing, stream processing, graph computation, and query analysis. Noteworthy within this spectrum are influential products such as MapReduce, Spark, Flink, and Hive, which have achieved widespread adoption. Generally speaking, big data technology provides effective solutions for the comprehensive management and rapid processing of large urban datasets. Moreover, it serves as a conduit for unearthing concealed knowledge within colossal datasets, thereby substantially bolstering urban applications across multiple domains [95].

4.3 Impact of cloud computing on USCS

Cloud computing represents a transformative paradigm in the realm of information technology, offering cost-effective, dynamically scalable computing services delivered through the Internet [96]. Its purpose is to cater to the user's specific needs, ensuring both expeditious and secure access to a plethora of services and data storage [97]. At its core, cloud computing involves the conversion of physical infrastructure, including network devices, high-performance servers, storage systems, and various software applications, into readily accessible online services, effectively disentangling the ownership of physical assets from the rights of usage [98]. In contrast to traditional computing models, cloud computing stands out with its distinctive features: resource pooling, widespread network accessibility, on-demand service, and rapid scalability. These inherent characteristics unlock substantial opportunities for optimizing service delivery and facilitating the efficient sharing of computing resources [99].

Currently, cloud computing has found extensive application across diverse domains of societal production, leading to significant advancements in government transformation, the sharing of educational resources, and the advancement of informatization [100]. Cloud computing harnesses the potential of cloud service platforms to furnish the public with data resources, computational capabilities, and specialized computing environments tailored to specific application scenarios [101, 102]. Moreover, cloud service platforms are not limited to offering universally applicable resources and environments; they also cater to the requirements of intelligent research domains

by providing specialized cloud computing services. Take remote sensing cloud services, for instance, where platforms such as Google Earth Engine and PIE-Engine Studio have garnered substantial audiences, effectively driving the open sharing of data products, and algorithms and expanding the reach of remote sensing applications [103]. The convenience of seamless online interactive computational analysis is driving the popularization of remote sensing applications.

4.4 Impact of AI on USCS

AI is an emerging scientific discipline dedicated to the exploration, formulation, and advancement of theories, methodologies, technologies, and practical systems aimed at emulating, extending, and augmenting human intelligence [104]. It represents one of the most dynamic and evolving domains in contemporary technology, encompassing a diverse range of research domains such as robotics and natural language processing, among others [105]. By amalgamating frontier technologies, AI not only empowers traditional industries but also exhibits a symbiotic relationship with them, fostering a synergistic co-evolution [106].

The outlook for the future advancement of AI is characterized by its remarkable expansiveness, with an increasingly broadening scope of applications within urban settings. This is most prominently exemplified by the emergence of smart cities and intelligent transportation systems, wherein urban intelligent applications are devised to achieve comprehensive coordination and management through intelligent methodologies [107]. AI has become essential to smart cities, significantly improving urban governance efficiency, enhancing citizens' quality of life, and fostering sustainable urban development [108]. Facilitated by cutting-edge technologies like big data and the IoT, AI-driven computational analyses of urban data enable more precise perception, prediction, and early warning systems pertaining to critical infrastructure and social security trends [109, 110]. The imperative to promptly integrate AI-driven computational processes into USCS underscores both its critical necessity and its potential for profound impact [111].

5 Services and applications

In addition to its role in aggregating and effectively managing a wide spectrum of urban data, an integrated USIS plays a pivotal role in providing a variety of application services across distinct urban domains, as visually represented in Fig. 6. Some indispensable services primarily encompass data sharing, visual analysis, intelligent computation, and decision support. This paper is dedicated to a comprehensive examination of select core services, emphasizing the system's relevance within the context of urban development and its multifaceted applications across different domains.

5.1 Data visualization

With the rapid advancement of geospatial big data technology, there has been an increasing emphasis on both the authenticity and visual appeal of digital information

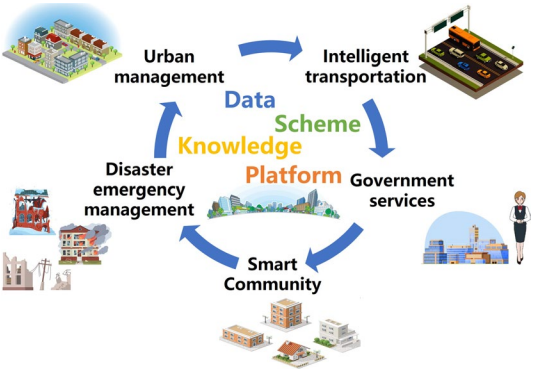


Fig. 6 Application services cases of the USCS

[112]. Consequently, there has been a gradual shift from conventional two-dimensional representations towards more intricate three-dimensional and multi-dimensional visualizations [113]. Notable progress has been achieved in transitioning from traditional two-dimensional vectors and rasters to sophisticated three-dimensional models, enabling computing devices to simulate urban scenarios across various spatial scales [114]. In the contemporary landscape, data visualization transcends mere data presentation and serves as a pivotal tool for spatial cognition, facilitating advanced analyses, and unveiling intricate relationships and patterns within data variations [115]. USCS face the significant challenge of enabling users to comprehend data easily and discover knowledge effectively through visualization. To this end, it is essential to present various types and dimensions of data in visually appealing and intuitive formats, while also promoting interactive exploration during visual analysis [116]. The evolution of technologies and theories in different fields has catalyzed the development of multidimensional data visualization techniques, such as two-dimensional flat representations, three-dimensional stereoscopic displays, spatiotemporal sequences, virtual reality, street view maps, and numerous other innovative methods [117].

A comprehensive data visualization platform for urban big data is crucial to fully leverage the potential of data and facilitate urban-related analysis and decision-making [118]. The platform must meet several essential criteria: firstly, it should be capable of accommodating vast amounts of spatiotemporal geographic data while maintaining high scalability. Secondly, it needs to enhance spatiotemporal data retrieval efficiency, reduce redundant data, and optimize data reusability and operational efficiency. Lastly, the platform must offer advanced analytical functionalities to meet users' demands for precise data analysis and decision-making.

5.2 Intelligent processing

The advancement of AI technology has ushered in a profound transformation in the comprehensive analysis and utilization of urban data, yielding substantial advantages across a spectrum of domains. The advancement of IoT and big data technologies facilitates the prolific generation of voluminous data, necessitating the expeditious extraction of latent patterns and insights from urban datasets. These insights are crucial for guiding urban management and simulation evaluations [119]. Similarly, the

forthcoming iteration of USCS is anticipated to exhibit heightened intelligence, tailored to meet the multifaceted service requisites.

In the current paradigm, the handling of urban spatial data necessitates the seamless integration of computational techniques from the realm of AI. This integration encompasses the deployment of sophisticated deep learning models for tasks such as urban functional zone extraction, land cover classification, intelligent point cloud data modeling, and environmental assessment [120]. To meet the evolving requirements of a diverse user base, USCS must be equipped with advanced processing capabilities rooted in AI technologies. This entails a comprehensive management of AI models, including machine learning and deep learning algorithms. Furthermore, it is imperative to underscore the significance of employing well-suited computational processing engines that align seamlessly with these intelligent models, all while prioritizing system scalability and computational efficiency.

5.3 Data sharing

Based on the USCS infrastructure, data sharing is facilitated across a multitude of domains, departments, and user groups, thereby unlocking its inherent potential [121]. To align with the prevailing trends in societal development and the construction of smart cities, it is imperative to offer extensive data-sharing services to the public. Realizing data openness and sharing involves a three-step process: aggregation and organization of diverse urban data from multiple sources, archiving this data into a database management system, and finally, providing users with comprehensive data views, flexible data retrieval, and data access through the system's client-side [122]. The USCS facilitates the seamless integration and efficient management of multiple urban data while providing essential data support to users and specific application services.

5.4 Decision support

In addition to facilitating data sharing, USCS should offer diverse computational analyses tailored to various scenarios, allowing for knowledge discovery and pattern recognition that can support urban management, construction planning, and other decision-making processes [151, 152]. The progression of smart city development and the digital twin practice furnish a pivotal technical foundation, enabling real-time data accessibility, visual representation, and computational analysis within these spatial information systems. Furthermore, it is imperative for the USCS to possess rapid and intelligent computational processing capabilities, coupled with the ability to render three-dimensional data visualization, in order to bolster decision-making proficiency [132]. The integration of cutting-edge technologies, including digital twin, visual analysis, and intelligent computation processing, will undoubtedly result in a marked enhancement of the system's efficiency and precision, augmenting its capacity to provide decision-making support across critical domains such as urban emergency management and disaster prevention.

5.5 Various applications

Leveraging USCS enables deep integration with specific business operations across various domains, facilitating corresponding computations, analyses, and services tailored to meet many application needs [153]. For instance, USCS can cater to business processes in areas such as government affairs, urban management, transportation, emergency response, healthcare, and smart communities [154].

We listed some applications based on USCS, mainly in information management, assessment analysis, smart city construction, etc., as shown in Table 3. For example, from the perspective of urban data management, Pang developed an information system for public transportation and community services, enabling users to share and access information based on location data [130]. In terms of urban assessment and analysis supported by GIS, Xu et al. identified vulnerable population and their precise cause of vulnerability by assessing flood hazard [131]. Furthermore, Jaydeep et al. explored many aspects of smart city construction based on GIS [145], accompanied

Table 3 Application cases based on USCS

Category	Specific Application	References
Data management	Urban data management	(Monteiro, 2018 [123]; Lv, 2016 [124]; Sularno, 2021 [125])
	Waste management	(Wu, 2016 [126]; Chabuk, 2016 [127]; Khan, 2016 [128])
	Government affairs	(Muslih, 2020 [129])
	Community management	(Pang, 2019 [130])
Assessment & Analysis	Disaster assessment	(Tanaya, 2020 [131]; Lee, 2018 [132])
	Greenbelt assessment	(Sembiring, 2018 [133])
	Living environment analysis	(Philia, 2023 [134])
	Groundwater quality	(Khan, 2017 [135]; Khosravi, 2017 [136]; Sargazi, 2021 [137]; Mohammadi, 2017 [138])
	Atmospheric concentration	(Ramahi, 2020 [139])
	Sewage treatment	(Zolfaghary, 2021 [140]
Urban construction	Smart transportation	(Gossling, 2016 [141]; Nadi, 2017 [142]; Parmar, 2020 [143]; Shafabakhsh, 2017 [144])
	Smart City	(Jaydeep, 2017 [145]; Jiang, 2020 [146])
	Construction Planning	(Kouziokas, 2016 [147]; Puchol, 2021 [148])
Others	Urban crime	(Cheng, 2021 [149])
	City security	(Mata, 2016 [150])

by various urban-related applications such as smart transportation and research of urban crime [141, 150].

6 Comparison of USCS

Currently, many cities are embarking on plans to develop smart cities, and some have already established digital urban management information platforms that integrate seamlessly with standard spatiotemporal data and services [7]. They provide a comprehensive solution for spatiotemporal data management and publishing, comprising a unified, cross-platform urban information model [155]. The presented examples of smart city construction cases aim to analyze the current technical and application status of existing urban spatial information systems. We selected several representative system platforms from different regions and countries with multiple application scenarios as cases to illustrate the current functional services and application status of USCS. It is hoped that readers will see the wide range of application scenarios and potential technology evolution paths of USCS from these platforms.

6.1 Visual Urban spatial digital platform

Shenzhen City has launched the Shenzhen Visualized Urban Spatial Digital Platform with the aim of modernizing the urban governance system and enhancing its capacity. The platform provides universal services for real-life three-dimensional space bases and spatial platforms throughout the entire city. It has successfully constructed China's first integrated land and sea three-dimensional spatiotemporal reference system, providing a foundation for precise perception of land-sea positions and supporting the integration of national spatial information resources with the Greater Bay Area. Presently, the platform offers general services, including geographic spatial data services, comprehensive space queries, and comprehensive visualization of urban space. Additionally, it has integrated a semantic address library with 12 million data records and has been piloted in fields such as national land spatial planning, social governance, intelligent transportation, and industrial park management.

Currently, the platform demonstrates a notable proficiency in the domains of data management and service sharing. Nevertheless, it exhibits certain limitations when it comes to offering comprehensive support for intelligent computation and analytical processing in intricate urban settings. Specifically, as a spatial data base, it centrally manages the city's two-dimensional vector data, three-dimensional building information models, as well as underground spatial data, such as underground pipelines and geological structures. In addition, it can also collect dynamic data such as traffic flow, environmental monitoring, and energy consumption around the city in real time. In terms of visualization, it can perform multi-source heterogeneous data fusion display, dynamic interaction and spatial analysis. At present, the platform has been put into in-depth application scenarios in sufficient districts and enterprises in Shenzhen, and will promote this solution in the international market. Overall, the digital twin platform built on frontier technologies, such as artificial intelligence, cloud computing, and edge computing, already has a variety of functions and processing and analysis

capabilities. However, there is still room for improvement in real-time data updates and blockchain applications.

6.2 UrbanPulse

UrbanPulse is an urban multi-source data platform that can aggregate and manage multi-scenario urban data collected by various sensors in real-time. Its goal is to provide an open urban information platform for smart city and community construction [156]. It offers intelligent services that provide users with access to various types of urban sensor data in a seamlessly integrated manner [157]. Figure 7 illustrates the system architecture of the platform, which employs a layered distributed overall design and is capable of achieving effective scalability of the cluster system. UrbanPulse has proven to be an effective solution for smart city construction and has been implemented in multiple cities. For instance, in the city of Bonn, the platform has been utilized to provide a range of application services, including sound pollution management, traffic management, smart parking, intelligent lighting, and smart pollution management.

As a spatial information platform for smart city clusters, UrbanPulse deeply integrates remote sensing, cloud computing and AI technologies. Its core advantages lie in its high-temporal and spatial resolution data processing capabilities and its scalable architecture that collaborates with the edge and cloud. Overall, UrbanPulse performs well in multi-source data management, visualization capabilities, and interactive analysis, and can provide a variety of functional services to the outside world. While UrbanPulse has made a notable contribution to the development of smart cities, it still possesses certain limitations. For instance, the urban data involved in this system is not abundant enough, as it exclusively caters to various sensor-generated datasets. If the platform could integrate a diverse range of urban data comprehen-

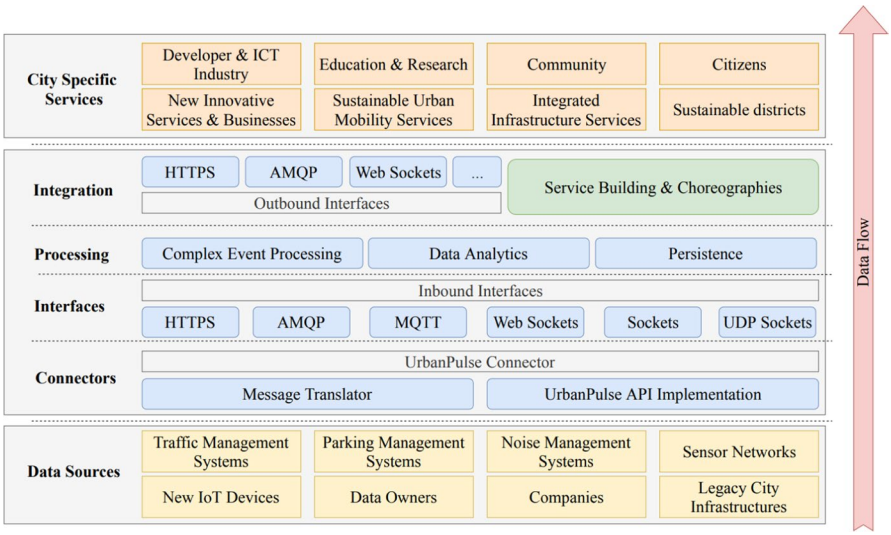


Fig. 7 The system architecture diagram of UrbanPulse

sively, it would better facilitate the advancement of smart city construction and many urban applications.

6.3 FIWARE open source platform

FIWARE is an open-source platform framework that facilitates the integration of various smart city solutions, thereby accelerating the development and deployment of related application systems [158]. The biggest feature of FIWARE is the degree of openness and interoperability. Currently, it has been applied to various aspects of smart city construction, such as global IoT market, smart cities, and autonomous driving. A software architecture or platform powered by FIWARE offers significant advantages in supporting the urban digital twin, enabling effective management of real-world digital twin data representation [159]. The Digital Twin data representation is constructed using information from diverse sources, including mobile devices, sensors, information systems, cameras, and social networks. Applications continuously process and analyze this data to automate tasks and facilitate informed decision-making for end users.

FIWARE's core data model is based on the next generation service interface standard and supports unified management and services of multi-source heterogeneous data. In terms of visualization, it supports integrated visualization of 2D maps and 3D models. Most importantly, it has good system expansion capabilities, a good community ecology, and high maturity. In contemporary times, platforms formulated upon the FIWARE architecture have garnered extensive utilization and have found application across numerous industries and research domains associated with urban settings. The research and application of the FIWARE framework are poised to become increasingly extensive in the future, exerting profound influence on the development of urban spatial information systems.

6.4 National Urban information system

To establish an urban geospatial database, India launched the National Urban Information System (NUIS) in 2006, aimed at mapping various towns using high-resolution remote sensing data. Additionally, to enhance the platform's accessibility and utilization, a web application named 'Bhuvan-NUIS' was developed. The Bhuvan platform manages the NUIS geographic information database, enabling users to access and leverage the stored data for analytical and planning purposes. Nonetheless, within the course of research and exploration, the NUIS still demands additional enhancements to accomplish its established objectives. Merely managing urban data falls short; the platform's scope should encompass a broader array of services, including enriched capabilities for computation, analysis, simulation, and assessment across diverse sectors.

Overall, the urban data that NUIS can manage is limited. The main reason is that the means of perceiving urban data are not comprehensive enough, resulting in few data sources that the system can access. The system has major problems with data privacy and security vulnerabilities. From the perspective of data processing, real-time processing capabilities are limited and AI applications lag. Although it supports

2D and 3D visualization, the dynamic rendering capabilities of complex scenes are insufficient. In contrast, compared with the aforementioned system cases, NUIS is less mature and its platform capabilities need to be improved. In the future, mainstream innovative technologies can expand the platform's functions and services.

7 System architecture design

In this section, we present an architectural design for USCS and concurrently develop a city remote sensing intelligent interpretation system based on this architecture as a system demonstration. Firstly, we propose a novel architectural design based on existing mainstream technologies for reference to researchers. Secondly, a system implementation serves as a demonstration of USCS, aiding in a better understanding of its functionalities and services.

7.1 A novel architecture design for USCS

An USCS is dedicated to aggregating and managing urban data, offering comprehensive data services, intelligent computation, and decision analysis applications for city management and planning. In light of this, we proposes a holistic outline architecture design for USCS based on established theoretical methodologies and cutting-edge technologies. The overall architecture is categorized into five layers based on the distinct functional characteristics at each level. These layers, arranged from the lowest to the highest, consist of the hardware infrastructure layer, data storage layer, processing engine layer, functional support layer, and application service layer. Figure 8 presents a detailed illustration of this architectural design. Each layer in the system serves a unique purpose, offering specific functional support. Next, we will delve into each layer, exploring its primary components, the key technologies it utilizes, and the support it extends to the system.

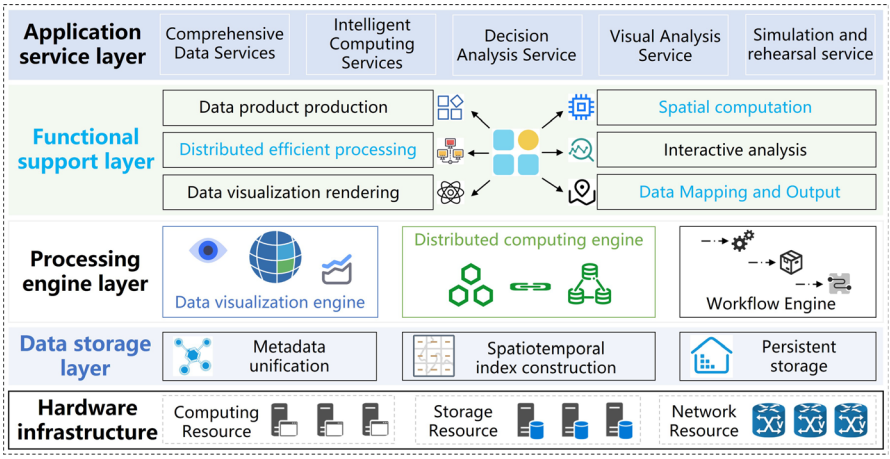


Fig. 8 A hierarchical architectural design for USCS

Hardware infrastructure layer: The construction of USCS requires several categories of fundamental hardware resources, such as storage space, network resources, and computing power. A prevalent approach involves leveraging high-performance server clusters to provide robust hardware support, capable of handling extensive storage and processing requirements for vast urban datasets. Additionally, the adoption of servers equipped with multi-core processors and high-performance graphics processing unit (GPU) significantly amplifies the cluster's data processing capacity. The ongoing surge in cloud computing and big data technologies has enabled server clusters to adopt elastic resource allocation and utilization, thus facilitating agile adaptation to the specific hardware resource demands of various systems. Relying on the flexible expansion and contraction of distributed clusters, the hardware infrastructure can be efficiently expanded to meet the ever-changing needs of a wide range of users and applications.

Data storage layer: A robust data organization and storage system is essential to handle the extensive data volumes in complex urban environments. It should enable the archival storage of various urban spatial data and provide data querying and access capabilities. Presently, distributed file systems or database systems such as HDFS, Ceph, ES, and others offer viable options for achieving data persistence. The distributed data management solutions have showed commendable performance in handling extensive data and ensuring data redundancy, earning them widespread adoption across various research domains. The storage process typically involves tasks such as unifying metadata, constructing spatiotemporal indices, and enabling persistent storage, ultimately facilitating comprehensive data access for upper-level applications or users. For data with time attributes, we use ES to manage metadata fields and create indexes, which can provide efficient queries for any time range or time point. As for different spatiotemporal data, the final storage solution can be selected according to their characteristics. Cloud computing technology can support the elastic allocation of storage space, and combined with big data management technology, it can support the provision of massive time series data storage and data discovery.

Processing engine layer: In addition to organizing and managing urban data, a USCS must offer diverse calculation and analysis capabilities to different users. The effective combination of data and processing applications is vital for fully harnessing the potential of the USCS and creating substantial value. The primary objective of the processing engine layer is to establish engines for computation and analysis, encompassing data visualization, intelligent computation, and workflow processing engines. By coherently integrating various processing engines, it furnishes essential support for intelligent computing, visual analysis, and interactive processing at higher levels. Currently, prevalent processing engines include Spark computing engine, container-based model computing engine, and user-centric data visualization engine, etc. Combined with mainstream technologies such as cloud computing and edge computing, it can provide flexible computing resource expansion for the processing engine and improve processing efficiency in complex scenarios. Especially for time series data, edge devices can complete fast processing near the data side, as well as efficient data computing and analysis based on stream data processing frameworks, such as Flink.

Functional support layer: USCS needs to cater to diverse functionalities across different application domains and departments. Apart from facilitating comprehensive data services, the systems must support multiple computational and analytical applications in various scenarios. Achieving this requires the logical implementation of multiple computational and analytical modules, including data product generation, spatial computation, distributed efficient processing, interactive analysis, data visualization, and cartographic output. By ensuring a holistic functional design and logical implementation, these modules effectively provide essential support to upper-level applications. In addition, spatiotemporal data desensitization algorithms and privacy-enhancing computing technologies can be combined to support the protection of privacy data. The above various calculation and analysis modules can all be managed as an algorithm library, and support customized extensions and calls, providing rich operator support for the system.

Application service layer: The primary objective of the layer is to offer users diverse computational analyses and support various application scenarios. These services encompass a wide range of offerings, including urban data services, intelligent computational analysis services, visualization analysis services, deduction services, and decision analysis. The application service layer directly engages with users, providing ample services that effectively support various urban applications, such as management, governance, planning, and development. The services provided by the system are directly user-oriented, including spatio-temporal data query and visual analysis, etc., and also need to be supported by a good interactive system interface to better serve users. In addition, based on the idea of microservices, the service side can be extended according to the real-time needs of users to improve the service capabilities and robustness of the system.

The architectural design proposed in this paper offers scalability potential, it can adapt to many different technical solutions and can apply to USCS research in many fields related to cities. First, the hardware expansion can be reflected in the hardware infrastructure layer, realizing the flexible expansion of servers and other equipment. In addition, relying on distributed clusters, the elastic expansion and allocation of storage space and computing power can be completed, supporting the needs of data storage and efficient computing at the data storage layer and processing engine layer, respectively. Meanwhile, the scalability of the architecture we proposed is also reflected in the application service layer, using microservice technology to achieve multi-terminal deployment on the service side. The development of the urban spatial information application system based on the system architecture designed in this paper can realize the access and management of multi-source urban spatial big data. Combining innovative technologies, such as cloud computing, remote sensing data processing, and AI, can enrich the system's functions and processing capabilities, and adapt to the needs of diverse urban application scenarios with good system scalability. The most important issue is to provide a user interface with flexible interaction and convenient services on the client side to achieve visible, accessible, and efficient services for data and computing. We believe that this architectural design schema will help solve the problems of incomplete design and unclear functional division that exist during the development and implementation of USCS.

7.2 System demonstration of USCS

In response to the demand for urban spatial information data sharing and land feature interpretation analysis, we developed a urban remote sensing intelligent interpretation system. The developed system serves as a prototype for USCS, showcasing its functions and services through a practical example. Subsequently, we will discuss its functions, technical components, and interface display.

The primary objective of the system is to delve into the management, processing, and service aspects of urban spatial data, with a particular emphasis on providing intelligent interpretation of remote sensing data. As illustrated in Fig. 9, it delineates the primary functionalities offered by the USCS demonstration system. First, the system centrally manages a variety of urban-related data, encompassing geographic feature data, remote sensing image data, and building model data, the total data volume reaches TB level. Second, we provide functionalities for spatial data processing, including vector-raster data processing and coordinate transformation. The system integrates multiple models to facilitate intelligent algorithm-driven spatial data processing and analysis, including over ten intelligent computing methods based on deep learning and numerical simulation analysis algorithms. Furthermore, the system can externally provide many services, including data sharing, interpretation processing, and visual analysis. In terms of visualization, it supports 2D data such as vectors, rasters, and charts, and can also efficiently render large-scale 3D building and terrain data. To achieve the above functions, we completed the system development on several Dell high-performance server based on programming languages/frameworks/components such as Java, Springboot, Cesium, and Vue. Among them, in order to develop a platform with good functions and performance, the main development stack uses Pytorch, GeoTrellis, Spark, Redis, MinIO, HBase, ElasticSearch and Flink.

The following aspects represent the key technologies involved in the development and implementation of this system. In terms of hardware infrastructure, we employed multiple high-performance servers to furnish the requisite hardware support for data storage and processing. We then built a distributed cluster that includes big data storage and processing components. On the one hand, utilizing distributed file systems and object storage enables centralized management of multi-source urban spatial data. On the other hand, leveraging the Spark processing engine and digital Earth visualization engine facilitates efficient processing and rendering of various types of

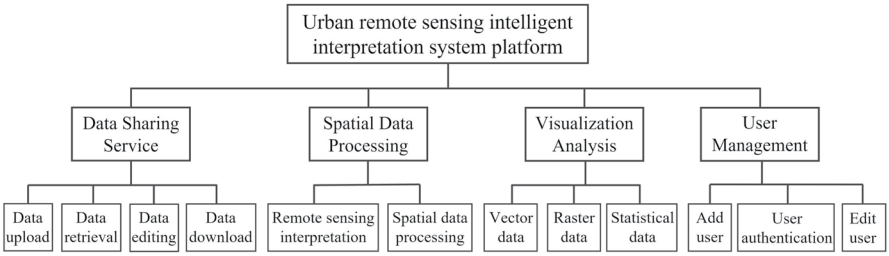
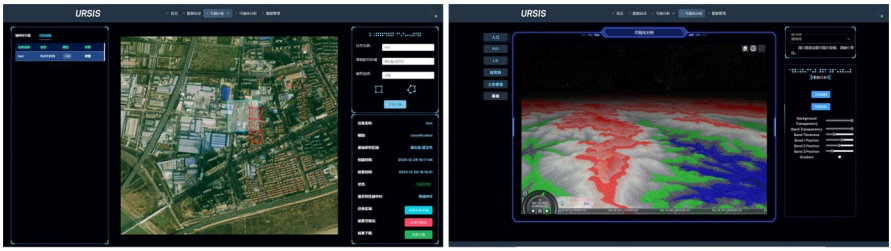


Fig. 9 The main functional modules of the demonstration system



(a) (b)

Fig. 10 The interface display of the demonstration system

spatial big data. At the functional support layer, we developed distinct spatial data processing modules, encompassing raster data slicing, visual analysis, and interpretation processing. Lastly, users are empowered with urban spatial data retrieval and download, interpretation processing and analysis, as well as visualization services. Overall, the system performs well in terms of system performance and interactive operation efficiency, and a variety of efficient processing and analysis provide a solid platform support for multi-scenario decision making and evaluation of urban space.

Data service sharing, associated analysis processing, and visual representation are all crucial functionalities for USCS. Here, we offer a concise depiction of two system interfaces as shown in 10. Figure 10 (a), it is an interface for urban functional zone interpretation and processing. This module provides users with intelligent interpretation and analysis of functional zones within custom spatial ranges. Figure 10 (b) represents the interface for visualizing multi-category spatial data in the system, demonstrating the visual rendering effects of elevation data here. Altogether, through interactive operations, our system offers users more flexible and convenient services for processing and analyzing urban spatial data.

8 Future directions

There are still many difficulties to be overcome and improved in the process of system construction and service applications, here we list a few specific issues that need to be addressed within the scope of this paper.

- Enable the USCS to achieve more efficient urban spatial data aggregation and management. As the means of acquiring data become increasingly diverse, efficient aggregation and management of data have become crucial [160]. It can establish a solid foundation for data-driven analysis and computation by integrating data from various scenes in urban spaces. Additionally, the organization of urban-related data must be more holistic and efficient, particularly regarding accessing and rapidly organizing vast amounts of real-time data, to effectively support users and various applications through proficient data management [161].
- Compatible with multiple intelligent computing models in a highly scalable form. The USCS not only can handle various types of data but also requires access to

multiple computational and analytical models for different data and application scenarios [162]. With the flourishing development of AI technology, the system must possess the ability to perform intelligent computation, analysis, and reasoning. This necessitates the creation of a processing engine that can seamlessly integrate intelligent computational models and provide users with intelligent computation and reasoning services [163]. Additionally, the system's scalability is of utmost importance, requiring the effortless introduction of constantly evolving computational models.

- To provide customized services with high interactivity and usability for users in multiple urban domains. An excellent USCS should encompass a diverse set of functions while also ensuring convenient and flexible operation. High availability and interactivity in a system can attract a broader user base, enhance its value, and increase its impact [164]. Moreover, USCS researchers should explore providing tailored services to meet specific needs in different industries during users' straightforward interactions.

9 Conclusion

This paper presents a comprehensive review of theories pertaining to USCS, exploring frontier technologies and their current applications. We begin by delineating the multiple data integral to USCS, explaining their sources and methodologies for acquisition. Subsequently, we delved into the data model underpinning urban spatial data, along with the database systems employed for its storage and management. Then, we discussed several frontier technologies that have significantly influenced the evolution of USCS, along with their applicability to USCS and the requisite services they cause. Concurrently, we scrutinize various extant USCS, offering illustrative examples, and proffer a system architecture design suitable as a reference for researchers engaged in related domains. In conclusion, we articulate a set of pertinent issues that merit further consideration in the future. Overall, this paper describes the current technical status and future development direction of USCS. Through an innovative system architecture design, it provides researchers with new ideas for technical research and application practice, and effectively promotes the construction of smart cities.

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Declarations

Conflict of interest The authors declare no Conflict of interest.

References

1. Supriadi A, Oswari T (2020) Analysis of geographical information system (gis) design aplication in the fire department of depok city. Technium Soc Sci J 8:1
2. Ma Z, Ren Y (2017) Integrated application of bim and gis: an overview. Proc Eng 196:1072–1079
3. Lü G et al (2019) Reflections and speculations on the progress in geographic information systems (gis): a geographic perspective. Int J Geogr Inf Sci 33:346–367
4. Silva BN, Khan M, Han K (2018) Towards sustainable smart cities: a review of trends, architectures, components, and open challenges in smart cities. Sustain Cities Soc 38:697–713
5. Wang L, Zuo B, Le Y, Chen Y, Li J (2023) Penetrating remote sensing: next-generation remote sensing for transparent earth. The Innovation 4:100519
6. Glaeser EL, Kominers SD, Luca M, Naik N (2018) Big data and big cities: the promises and limitations of improved measures of urban life. Econ Inq 56:114–137
7. Barns S (2018) Smart cities and urban data platforms: designing interfaces for smart governance. City Cult Soc 12:5–12
8. Letaifa SB (2015) How to strategize smart cities: revealing the smart model. J Bus Res 68:1414–1419
9. Rashid B, Rehmani MH (2016) Applications of wireless sensor networks for urban areas: a survey. J Netw Comput Appl 60:192–219
10. Zheng Y, Capra L, Wolfson O, Yang H (2014) Urban computing: concepts, methodologies, and applications. ACM Trans Intell Syst Technol (TIST) 5:1–55
11. Soomro K, Bhutta MNM, Khan Z, Tahir MA (2019) Smart city big data analytics: An advanced review. Wiley Interdiscip Rev Data Mining Knowl Discov 9:e1319
12. Allam Z, Dhunny ZA (2019) On big data, artificial intelligence and smart cities. Cities 89:80–91
13. Zantalis F, Koulouras G, Karabetsos S, Kandris D (2019) A review of machine learning and iot in smart transportation. Future Internet 11:94
14. Kaffash S, Nguyen AT, Zhu J (2021) Big data algorithms and applications in intelligent transportation system: a review and bibliometric analysis. Int J Prod Econ 231:107868
15. Meijer A, Bolívar MPR (2016) Governing the smart city: a review of the literature on smart urban governance. Int Rev Adm Sci 82:392–408
16. Andronie M, Lăzăroiu G, Iatagan M, Hurloiu I, Dijmărescu I (2021) Sustainable cyber-physical production systems in big data-driven smart urban economy: a systematic literature review. Sustainability 13:751
17. Wang C et al (2022) Data acquisition for urban building energy modeling: a review. Build Environ 217:109056
18. Yang B, Xu T, Shi L (2017) Analysis on sustainable urban development levels and trends in China's cities. J Clean Prod 141:868–880
19. Boeing G (2021) Spatial information and the legibility of urban form: big data in urban morphology. Int J Inf Manage 56:102013
20. Kaplan D, Holloway S Urban geography (John Wiley & Sons, 2024)
21. Althoff T et al (2017) Large-scale physical activity data reveal worldwide activity inequality. Nat 547:336–339
22. Kitchin R (2014) The real-time city? Big data and smart urbanism. GeoJournal 79:1–14
23. Gupta A, Panagiotopoulos P, Bowen F (2020) An orchestration approach to smart city data ecosystems. Technol Forecast Soc Chang 153:119929
24. Ajay P, Nagaraj B, Pillai BM, Suthakorn J, Bradha M (2022) Intelligent ecofriendly transport management system based on iot in urban areas. Environment, Development and Sustainability 1–8
25. Chen, N. et al. Dynamic urban surveillance video stream processing using fog computing (2016). 2016 IEEE second international conference on multimedia big data (BigMM), 105–112, IEEE
26. Toth C, Jóźków G (2016) Remote sensing platforms and sensors: a survey. ISPRS J Photogramm Remote Sens 115:22–36
27. Xue J (2017) Su B (2017) Significant remote sensing vegetation indices: a review of developments and applications. J Sens 1:1353691
28. Rathore MM et al (2018) Exploiting iot and big data analytics: defining smart digital city using real-time urban data. Sustain Cities Soc 40:600–610
29. Deilami K, Kamruzzaman M, Liu Y (2018) Urban heat island effect: a systematic review of spatio-temporal factors, data, methods, and mitigation measures. Int J Appl Earth Obs Geoinf 67:30–42
30. Dubovik O et al (2021) Grand challenges in satellite remote sensing. Front Remote Sens 2:619818

31. Roy P, Behera M, Srivastav S (2017) Satellite remote sensing: sensors, applications and techniques. Proc Natl Acad Sci, India, Sect A 87:465–472
32. Jonsson T (2023) Conceptual data systems architecture principles for information systems. Front Comput Sci 4:1008296
33. Elhorst JP, Elhorst JP (2014) Spatial panel data models. Spatial econometrics: From cross-sectional data to spatial panels 37–93
34. Singla S, Eldawy A, Diao T, Mukhopadhyay A, Scudiero E (2021) Experimental study of big raster and vector database systems. 2021 IEEE 37th International Conference on Data Engineering (ICDE), 2243–2248, IEEE
35. Baier G, Deschemps A, Schmitt M, Yokoya N (2021) Synthesizing optical and sar imagery from land cover maps and auxiliary raster data. IEEE Trans Geosci Remote Sens 60:1–12
36. Breunig M et al (2020) Geospatial data management research: progress and future directions. ISPRS Int J Geo Inf 9:95
37. Yao X, Li G (2018) Big spatial vector data management: a review. Big Earth Data 2:108–129
38. Benelli G, Duggan MF (2018) Management of arthropod vector data-social and ecological dynamics facing the one health perspective. Acta Trop 182:80–91
39. Liu J et al (2020) Urban big data fusion based on deep learning: an overview. Inform Fus 53:123–133
40. Wang S (2016) Cybergis and spatial data science. GeoJournal 81:965–968
41. Cressie N Statistics for spatial data (John Wiley & Sons, 2015)
42. Villarroya S, Viqueira JR, Cotos JM, Taboada JA (2022) Enabling efficient distributed spatial join on large scale vector-raster data lakes. IEEE Access 10:29406–29418
43. Liu C, Wu J, Kohli P, Furukawa Y (2017) Raster-to-vector: Revisiting floorplan transformation. Proceedings of the IEEE International Conference on Computer Vision, 2195–2203, IEEE
44. Homburg T, Staab S, Janke D (2020) Geosparql+: Syntax, semantics and system for integrated querying of graph, raster and vector data. The Semantic Web–ISWC 2020: 19th International Semantic Web Conference, Athens, Greece, November 2–6, 258–275, Springer
45. Sevgican S, Turan M, Gökarslan K, Yilmaz HB, Tugcu T (2020) Intelligent network data analytics function in 5g cellular networks using machine learning. J Commun Netw 22:269–280
46. Chaney T (2014) The network structure of international trade. Am Econ Rev 104:3600–3634
47. Atluri G, Karpatne A, Kumar V (2018) Spatio-temporal data mining: A survey of problems and methods. ACM Comput Surv (CSUR) 51:1–41
48. Ferreira N, Poco J, Vo HT, Freire J, Silva CT (2013) Visual exploration of big spatio-temporal urban data: A study of new york city taxi trips. IEEE Trans Visual Comput Graphics 19:2149–2158
49. Chen BY et al (2016) Spatiotemporal data model for network time geographic analysis in the era of big data. Int J Geogr Inf Sci 30:1041–1071
50. Wang S, Cao J, Philip SY (2020) Deep learning for spatio-temporal data mining: a survey. IEEE Trans Knowl Data Eng 34:3681–3700
51. Li J, Feng X, Yin J, Chen F (2020) Change analysis of spring vegetation green-up date in qinba mountains under the support of spatiotemporal data cube. J Sens 2020:1–12
52. Dathan B, Ramnath S Modeling Object-Oriented Systems (Springer, 2025)
53. Marron JS, Alonso AM (2014) Overview of object oriented data analysis. Biom J 56:732–753
54. Pahwa P, Chhabra R (2014) An object oriented data warehouse design. International Journal of Soft Computing and Engineering (IJSCE), IV (2) 2231–2307
55. Semnani NM, Kuper P, Breunig M, Al-Doori M (2018) Towards an intelligent platform for big 3d geospatial data management. ISPRS Ann Photogramm Remote Sens Spat Inf Sci 4:133–140
56. Hou W et al (2016) Assessing quality of urban underground spaces by coupling 3d geological models: the case study of foshan city, south china. Comput Geosci 89:1–11
57. Pan X, Chu J, Aung Z, Chiam K, Wu D (2020) 3d geological modelling: a case study for singapore Information Technology in Geo-Engineering: Proceedings of the 3rd International Conference (ICITG), Guimarães, Portugal 3, 161–167, Springer
58. Ran X, Xue L, Sang X, Pei Y, Zhang Y (2022) Intelligent generation of cross sections using a conditional generative adversarial network and application to regional 3d geological modeling. Math 10:4677
59. Ding C, Feng W, Li X, Zhou N (2019) Urban-scale building energy consumption database: a case study for Wuhan, China. Energy Proc 158:6551–6556
60. Mohamed MA, Altrafi OG, Ismail MO (2014) Relational vs. nosql databases: a survey. Int J Comput Inf Technol 3:598–601
61. Abedjan Z, Golab L, Naumann F (2015) Profiling relational data: a survey. VLDB J 24:557–581

62. Vyawahare H, Karde PP, Thakare VM (2018) A hybrid database approach using graph and relational database. 2018 International Conference on Research in Intelligent and Computing in Engineering (RICE), 1–4, IEEE
63. Groom CR, Bruno IJ, Lightfoot MP, Ward SC (2016) The cambridge structural database. Acta Crystallogr Section B Struct Sci Cryst Eng Mater 72:171–179
64. Zimányi E, Sakr M, Lesuisse A (2020) Mobilitydb: a mobility database based on postgresql and postgis. ACM Trans Database Syst (TODS) 45:1–42
65. Corbellini A, Mateos C, Zunino A, Godoy D, Schiaffino S (2017) Persisting big-data: The nosql landscape. Inf Syst 63:1–23
66. Ahmed MR, Khatun MA, Ali A, Sundaraj K (2018) A literature review on nosql database for big data processing. Int J Eng Technol 7:902–906
67. Guo D, Onstein E (2020) State-of-the-art geospatial information processing in nosql databases. ISPRS Int J Geo Inf 9:331
68. Raut A (2017) Nosql database and its comparison with rdbms. Int J Comput Intell Res 13:1645–1651
69. Rautmare S, Bhalerao D (2016) Mysql and nosql database comparison for iot application. 2016 IEEE international conference on advances in computer applications (ICACA), 235–238, IEEE
70. Vera-Olivera H et al (2021) Data modeling and nosql databases-a systematic mapping review. ACM Comput Surv (CSUR) 54:1–26
71. Meier A, Kaufmann M, Meier A, Kaufmann M (2019) Nosql databases. SQL & NoSQL Databases: Models, Languages, Consistency Options and Architectures for Big Data Management 201–218
72. Madison M, Barnhill M, Napier C, Godin J (2015) Nosql database technologies. J Int Technol Inf Manag 24:1
73. Zhang H, Chen G, Ooi BC, Tan K-L, Zhang M (2015) In-memory big data management and processing: a survey. IEEE Trans Knowl Data Eng 27:1920–1948
74. Choi JM, Jeong DW, Yoon JS, Lee SJ (2016) Digital forensics investigation of redis database. KIPS Trans Comput Commun Syst 5:117–126
75. Liu Q, Yuan H (2019) A high performance memory key-value database based on redis. J Comput 14:170–183
76. Wang K, Liu G, Zhai M, Wang Z, Zhou C (2019) Building an efficient storage model of spatial-temporal information based on hbase. J Spat Sci 64:301–317
77. Hassan MU, Yaqoob I, Zulfiqar S, Hameed IA (2021) A comprehensive study of hbase storage architecture—a systematic literature review. Symmetry 13:109
78. Tang X et al (2022) A ceph-based storage strategy for big gridded remote sensing data. Big Earth Data 6:323–339
79. LeFevre J, Maltzahn C (2020) Skyhookdm: Data processing in ceph with programmable storage. USENIX login;**45**
80. Fan Y, Wang Y, Ye M (2018) An improved small file storage strategy in ceph file system. 2018 14th International Conference on Computational Intelligence and Security (CIS), 488–491, IEEE
81. Ahir DD, Shaikh NF (2024) Evaluation of elasticsearch ecosystem including machine learning capabilities. Int J Safety & Secur Eng**14**(4)
82. Shah N, Willick D, Mago V (2022) A framework for social media data analytics using elasticsearch and kibana. Wireless Netw. 1–9
83. Bajer M (2017) Building an iot data hub with elasticsearch, logstash and kibana. 2017 5th International Conference on Future Internet of Things and Cloud Workshops (FiCloudW), 63–68, IEEE
84. Guia J, Soares VG, Bernardino J (2017) Graph databases: Neo4j analysis. ICEIS 1:351–356
85. Fernandes D, Bernardino J (2018) Graph databases comparison: Allegrograph, arangodb, infinitegraph, neo4j, and orientdb. Data 18:373–380
86. Ghazi MR, Gangodkar D (2015) Hadoop, mapreduce and hdfs: a developers perspective. Proc Comput Sci 48:45–50
87. Karun AK, Chitharanjan K (2013) A review on hadoop—hdfs infrastructure extensions. 2013 IEEE conference on information & communication technologies, 132–137, IEEE
88. Zhu L et al (2018) A review: remote sensing sensors. Multi-purposeful Appl Geospatial Data 19:19–42
89. Li J et al (2020) A review of remote sensing for environmental monitoring in China. Remote Sens 12:1130
90. Zou Q (2017) Research on cloud computing for disaster monitoring using massive remote sensing data. 2017 IEEE 2nd International Conference on Cloud Computing and Big Data Analysis (ICCCBDA), 29–33, IEEE

91. Ma Y et al (2015) Remote sensing big data computing: challenges and opportunities. Futur Gener Comput Syst 51:47–60
92. Sagiroglu S, Sinanc D (2013) Big data: A review. 2013 international conference on collaboration technologies and systems (CTS), 42–47, IEEE
93. Siddiqa A, Karim A, Gani A (2017) Big data storage technologies: a survey. Front Inf Technol Electron Eng 18:1040–1070
94. Cai H, Xu B, Jiang L, Vasilakos AV (2016) Iot-based big data storage systems in cloud computing: perspectives and challenges. IEEE Internet Things J 4:75–87
95. Assunção MD, Calheiros RN, Bianchi S, Netto MA, Buyya R (2015) Big data computing and clouds: trends and future directions. J Parallel Distrib Comput 79:3–15
96. Jamsa K Cloud computing (Jones & Bartlett Learning, 2022)
97. Srivastava P, Khan R (2018) A review paper on cloud computing. Int J Adv Res Comput Sci Softw Eng 8:17–20
98. Rashid A, Chaturvedi A (2019) Cloud computing characteristics and services: a brief review. Int J Comput Sci Eng 7:421–426
99. Rittinghouse JW, Ransome JF Cloud computing: implementation, management, and security (CRC press, 2017)
100. Kumar PR, Raj PH, Jelciana P (2017) Exploring security issues and solutions in cloud computing services-a survey. Cybern Inf Technol 17:3–31
101. Wang L, Ma Y, Yan J, Chang V, Zomaya AY (2018) pipscloud: High performance cloud computing for remote sensing big data management and processing. Futur Gener Comput Syst 78:353–368
102. Li J, Wang J, Yang L, Ye H (2022) Spatiotemporal change analysis of long time series inland water in sri lanka based on remote sensing cloud computing. Sci Rep 12:766
103. Amani M et al (2020) Google earth engine cloud computing platform for remote sensing big data applications: a comprehensive review. IEEE J Sel Topics Appl Earth Obs Remote Sens 13:5326–5350
104. Ertel W Introduction to artificial intelligence (Springer, 2018)
105. Chowdhary K Fundamentals of artificial intelligence (Springer, 2020)
106. Davenport TH, Ronanki R et al (2018) Artificial intelligence for the real world. Harv Bus Rev 96:108–116
107. Yigitcanlar T, Mehmood R, Corchado JM (2021) Green artificial intelligence: towards an efficient, sustainable and equitable technology for smart cities and futures. Sustainability 13:8952
108. Ashwini B, Savithramma R, Sumathi R (2022) Artificial intelligence in smart city applications: An overview. 2022 6th International Conference on Intelligent Computing and Control Systems (ICICCS), 986–993, IEEE
109. Yigitcanlar T, Cugurullo F (2020) The sustainability of artificial intelligence: an urbanistic viewpoint from the lens of smart and sustainable cities. Sustainability 12:8548
110. Herath H, Mittal M (2022) Adoption of artificial intelligence in smart cities: a comprehensive review. Int J Inf Manag Data Insights 2:100076
111. Voda AI, Radu LD (2018) Artificial intelligence and the future of smart cities. BRAIN Broad Res Artif Intell Neurosci 9:110–127
112. Perin C et al. (2018) State of the art of sports data visualization. Computer Graphics Forum, 663–686, Wiley Online Library
113. Qin X, Luo Y, Tang N, Li G (2020) Making data visualization more efficient and effective: a survey. VLDB J 29:93–117
114. Sadiku M, Shadare AE, Musa SM, Akujuobi CM, Perry R (2016) Data visualization. Int J Eng Res And Adv Technol (IJERAT) 2:11–16
115. Ware C Information visualization: perception for design (Morgan Kaufmann, 2019)
116. Unwin A (2020) Why is data visualization important? What is important in data visualization? Harvard Data Sci Rev 2:1
117. Alsallakh B et al (2016) The state-of-the-art of set visualization. Computer Graphics Forum, 35, 234–260, Wiley Online Library
118. Cepero T, Montané-Jiménez LG, Toledo-Toledo G (2021) Visualization technologies to support decision-making in city management. Program Comput Softw 47:803–816
119. Xiang X, Li Q, Khan S, Khalaf OI (2021) Urban water resource management for sustainable environment planning using artificial intelligence techniques. Environ Impact Assess Rev 86:106515
120. Ullah Z, Al-Turjman F, Mostarda L, Gagliardi R (2020) Applications of artificial intelligence and machine learning in smart cities. Comput Commun 154:313–323

121. Chawinga WD, Zinn S (2019) Global perspectives of research data sharing: a systematic literature review. Library Inf Sci Res 41:109–122
122. Welch EW, Feeney MK, Park CH (2016) Determinants of data sharing in us city governments. Gov Inf Q 33:393–403
123. Monteiro CS, Costa C, Pina A, Santos MY, Ferrão P (2018) An urban building database (ubd) supporting a smart city information system. Energy Build 158:244–260
124. Lv Z et al (2016) Managing big city information based on webvrgis. IEEE access 4:407–415
125. Sularno S, Astri R, Anggraini P, Mulya DP, Mulya D (2021) Geographical information system of bus and travel counter in padang city using bfs method based on mobile web. Sci J Inf 8:304–313
126. Wu H et al (2016) An innovative approach to managing demolition waste via gis (geographic information system): a case study in shenzhen city, china. J Clean Prod 112:494–503
127. Chabuk A, Al-Ansari N, Hussain HM, Knutsson S, Pusch R (2016) Landfill site selection using geographic information system and analytical hierarchy process: a case study al-hillah qadhaa, babylon, iraq. Waste Manag Res 34:427–437
128. Khan D, Samadder S (2016) Allocation of solid waste collection bins and route optimisation using geographical information system: a case study of dhanbad city, india. Waste Manag and Res 34:666–676
129. Muslih I, Nurdin N, Marzuki M (2020) Effectiveness of marriage services through information system management (simkah) at palu city religious court. Int J Contemp Islamic Law Soc 2:20–35
130. Pang C, Pan R, Neustaedter C, Hennessy K (2019) City explorer: The design and evaluation of a location-based community information system. Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems, 1–15
131. Sarmah T, Das S, Narendr A, Aithal BH (2020) Assessing human vulnerability to urban flood hazard using the analytic hierarchy process and geographic information system. Int J Disaster Risk Reduct 50:101659
132. Lee S, Lee S, Lee M-J, Jung H-S (2018) Spatial assessment of urban flood susceptibility using data mining and geographic information system (gis) tools. Sustainability 10:648
133. Sembiring IP, Batubara R, Patana P et al. (2018) Mapping of tree damage classification in the western part of medan city green belts using geographic information system. IOP Conference Series: Earth and Environmental Science, 166, 012–020, IOP Publishing
134. Latue PC, Rakuasa H (2023) Spatial analysis of landscape suitability of ambon city for settlement using geographic information system. Jurnal Riset Multidisiplin Dan Inovasi Teknologi 1:59–69
135. Khan R, Jhariya D (2017) Groundwater quality assessment for drinking purpose in raipur city, chhattisgarh using water quality index and geographic information system. J Geol Soc India 90:69–76
136. Khosravi R et al (2017) Use of geographic information system and water quality index to assess groundwater quality for drinking purpose in birjand city, iran. Desalin Water Treat 67:74–83
137. Sargazi S et al (2021) The application of geographical information system (gis) approach for assessment of groundwater quality of zahedan city, sistan and baluchestan province, iran. Groundw Sustain Dev 12:100509
138. Mohammadi AA et al (2017) Temporal and spatial variation of chemical parameter concentration in drinking water resources of bandar-e gaz city using geographic information system. Desalin Water Treat 68:170–176
139. Al-Ramahi FK (2020) Spatial analysis of radon gas concentration distrbuted at baghdad city using remote sensing and geographic information system techniquesd. Iraqi J Agricultural Sci **51**
140. Zolfaghary P, Zakerinia M, Kazemi H (2021) A model for the use of urban treated wastewater in agriculture using multiple criteria decision making (mcdm) and geographic information system (gis). Agric Water Manag 243:106490
141. Gössling S, Schröder M, Späth P, Freytag T (2016) Urban space distribution and sustainable transport. Transp Rev 36:659–679
142. Nadi PA, Murad A et al (2017) Reviewing the use of geographic information system (gis) to measure sustainable urban transport performance. J Geosci Eng Environ Technol 2:171–177
143. Parmar J, Das P, Dave SM (2020) Study on demand and characteristics of parking system in urban areas: A review. J Traffic Transp Eng (English Edition) 7:111–124
144. Shafabakhsh GA, Famili A, Bahadori MS (2017) Gis-based spatial analysis of urban traffic accidents: Case study in mashhad, iran. J Traffic Trans Eng (English edition) 4:290–299
145. Lella J, Mandla VR, Zhu X (2017) Solid waste collection/transport optimization and vegetation land cover estimation using geographic information system (gis): A case study of a proposed smart-city. Sustain Cities Soc 35:336–349

146. Jiang D (2020) The construction of smart city information system based on the internet of things and cloud computing. Comput Commun 150:158–166
147. Kouziokas GN (2016) Geospatial based information system development in public administration for sustainable development and planning in urban environment. Euro J Sustain Dev 5:347–347
148. Puchol-Salort P, O'Keeffe J, van Reeuwijk M, Mijic A (2021) An urban planning sustainability framework: systems approach to blue green urban design. Sustain Cities Soc 66:102677
149. Cheng T, Chen T (2021) Urban crime and security. Urban Informatics 213–228
150. Mata F et al (2016) A mobile information system based on crowd-sensed and official crime data for finding safe routes: a case study of mexico city. Mobile Inf Syst 2016(1):8068209
151. Hunter F Community power structure: a study of decision makers (UNC Press Books, 2017)
152. Gessa A, Sancha P (2020) Environmental open data in urban platforms: an approach to the big data life cycle. J Urban Technol 27:27–45
153. Eicker U, Weiler V, Schumacher J, Braun R (2020) On the design of an urban data and modeling platform and its application to urban district analyses. Energy and Build 217:109954
154. Schrotter G, Hürzeler C (2020) The digital twin of the city of zurich for urban planning. PFG-J Photogramm Remote Sens Geoinformation Sci 88:99–112
155. Haveri A, Anttiroiko A-V (2023) Urban platforms as a mode of governance. Int Rev Adm Sci 89:3–20
156. Miranda F et al (2016) Urban pulse: capturing the rhythm of cities. IEEE Trans Visual Comput Graphics 23:791–800
157. Agnihotri P, Luthra M, Peters S (2019) Urbanpulse: Adaptable middleware to offer city and user centric smart city solution. Proceedings of the 20th International Middleware Conference Demos and Posters, 29–30
158. Cirillo F et al (2019) A standard-based open source iot platform: Fiware. IEEE Internet of Things Magazine 2:12–18
159. Fazio M, Celesti A, Márquez FG, Glikson A, Villari M (ISCC) Exploiting the fiware cloud platform to develop a remote patient monitoring system (2015). 2015 IEEE symposium on computers and communication, 264–270, IEEE
160. König PD (2021) Citizen-centered data governance in the smart city: from ethics to accountability. Sustain Cities Soc 75:103308
161. Kontokosta CE (2021) Urban informatics in the science and practice of planning. J Plan Educ Res 41:382–395
162. Nikitas A, Michalakopoulou K, Njoya ET, Karampatzakis D (2020) Artificial intelligence, transport and the smart city: definitions and dimensions of a new mobility era. Sustainability 12:2789
163. Lv Z, Chen D, Lou R, Wang Q (2021) Intelligent edge computing based on machine learning for smart city. Futur Gener Comput Syst 115:90–99
164. Cugurullo F (2020) Urban artificial intelligence: from automation to autonomy in the smart city. Front Sustain Cities 2:38

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